ChangeTrace

A Change Evidence Chain for construction paperwork

REQUEST > APPROVAL > INVOICE

01 What does the Skill do?

ChangeTrace proposes a read-only review linking a construction change request, approval record, and invoice charge through inspectable evidence.

02 Who is the target user and what is their problem?

Project managers and subcontractors often reconcile scattered change paperwork manually; a charge can appear consistent even when its approval, scope, or billing basis is unclear.

03 What is the proposed solution?

Build a Change Evidence Chain: retain each selected document as a distinct source, surface supported identifiers and amounts, and flag gaps or conflicts for human review.

04 What is the core user flow?

Select one request, one approval, and one invoice. Inspect each path, analyze each accessible document separately, show source-linked observations, and compare only supported, comparable amounts. Otherwise explain what is missing.

05 How is it different from existing tools or conventional AI?

Rather than producing a free-form summary, the proposed chain requires evidence for material observations and withholds unsupported comparisons. Specialized construction platforms also address change controls; this is a narrow NAS-based review workflow, not a claim of category novelty.

Technical approach & validation

An early build with explicit evidence and scope gates

06 Which UGREEN AI NAS capabilities are used, and when?

First, file-agent uses ug-filesystem stat to inspect each selected path. Then document-agent uses ug-document qa or extract to analyze each selected document and request inspectable evidence where available, while reporting coverage limitations.

07 What is the technical approach?

The parent delegates self-contained tasks to registered experts. Document tasks specify one selected file and a query. Access must be authorized and established before reading. The selected actions (`stat`, `qa`, and `extract`) are read-only; no save or transfer actions are requested. Effective runtime permissions remain unverified. No saves or transfers are requested. Model choice and data locality depend on the UGREEN configuration and remain unverified.

08 What is built, unbuilt, and pending NAS testing?

A Markdown orchestration Skill has passed a static checker obtained from a third-party mirror; the mirror's official provenance has not been independently verified. Six synthetic PDFs were locally text-checked. No NAS workflow, expert extraction, source attribution, or single-file enforcement has been verified. No dashboard or runtime integration has been built.

09 How will the Skill be tested?

Core 1: with supported evidence and comparable full-charge terms, compare a USD 1,500 invoice with a USD 1,200 approval and inspect the USD 300 difference. Core 2: a progress charge with unclear tax and prior billing must not be subtracted. Edge: two explicitly selected conflicting approvals stay separate and require human resolution.

10 What are the limitations, risks, and dependencies?

The UGREEN environment and expert behavior are dependencies. PDF support, source evidence, scope enforcement, arithmetic, permissions, and model data handling are unproven. No result establishes legal approval, signature authenticity, completeness, payment entitlement, or guaranteed accuracy.

DEVELOPMENT DISCLOSURE

Development aids: Arena.ai Agent Mode (drafting and technical review); supplied UGREEN creation-tool references and mirror checker (github.com/sirmos/heirloom; mirror origin and license not independently verified); ReportLab 4.4.9 (BSD 3-Clause, reportlab.com) for this PDF; pypdf 6.4.0 (BSD 3-Clause, github.com/py-pdf/pypdf) for local fixtures. No third-party component or external service is bundled or requested by the Skill.